

Financial Performance Analysis

Budget & Expense Analysis

PREPARED BY

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TOOLS & TECHNOLOGIES

Microsoft Excel | Oracle SQL | Tableau

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Abstract

This project focuses on analyzing financial transaction data to understand the company's budget and expense performance and identify useful spending patterns. The main objective was to compare planned budgets with actual expenses and convert the financial data into meaningful business insights.

The project was carried out using Microsoft Excel, Oracle SQL, and Tableau. The dataset was first reviewed and prepared to ensure that the data was suitable for analysis. Excel was used for KPI calculations, statistical analysis, Pivot Tables, Pivot Charts, and dashboard creation. SQL was used to perform the defined business analysis, while Tableau was used to create an interactive dashboard for presenting the findings clearly.

The analysis showed that actual expenses were higher than the planned budget, with differences observed across months, departments, categories, and regions. The results helped identify major spending areas and highlight budget gaps that can be useful for financial monitoring and decision-making.

Overall, this project provided practical experience in working with financial data, performing analysis using multiple tools, creating dashboards, and presenting business insights. It also helped strengthen my analytical and technical skills as an aspiring Data Analyst.

Introduction

Companies generate financial data through their daily activities, and analyzing this data helps them monitor expenses, compare planned budgets with actual spending, and identify areas that need attention. Data analysis can help turn this financial information into useful insights for better business decisions.

This project focuses on analyzing financial transaction data to understand the company's budget and expense performance using Oracle SQL, Microsoft Excel, and Tableau. The analysis covers data validation, KPI calculation, business analysis, statistical analysis, and dashboard creation to identify spending patterns across different months, departments, categories, and regions.

Objectives

- To compare budgeted expenses with actual expenses.
- To identify monthly spending patterns and budget gaps.
- To analyze budget and actual expenses across departments.
- To identify the major categories contributing to actual expenses.
- To understand the distribution of expenses across different regions.
- To present financial performance through clear dashboards and insights.

Dataset Information

Dataset Information	Details
Dataset Name	Financial performance budget Dataset
Source	Kaggle (Public Dataset)
Total Records	10,000
Total Columns	9
File Format	CSV

Dataset Description

The dataset contains 10,000 financial transaction records with details such as transaction date, department, category, region, budget amount, actual amount, payment method, transaction ID, and month-year information. It was used to compare budgeted expenses with actual spending and to understand expense patterns across different periods, departments, categories, and regions.

Dataset Preview

Date	Department	Category	Region	Budget Amount	Actual Amount	Payment Method	Transaction ID
2023-05-11 00:00:00	Sales	Travel	North	126096	43048	Card	TXN100000
2023-11-11 00:00:00	Marketing	Salaries	East	19702	87896	Bank Transfer	TXN100001
2021-05-02 00:00:00	IT	Training	Central	108523	103632	Card	TXN100002
2022-04-12 00:00:00	Marketing	Salaries	North	114711	105574	Cash	TXN100003
2021-11-27 00:00:00	Sales	Utilities	Central	121895	64314	UPI	TXN100004
2021-03-29 00:00:00	Finance	Marketing	East	37762	77044	Bank Transfer	TXN100005
2023-05-22 00:00:00	IT	Marketing	West	144311	69127	UPI	TXN100006
2021-05-11 00:00:00	HR	Marketing	Central	22567	53015	Cash	TXN100007
2023-02-09 00:00:00	IT	Infrastructure	West	95737	58734	UPI	TXN100008
2021-12-10 00:00:00	IT	Salaries	East	16857	153432	Cash	TXN100009
2023-03-17 00:00:00	IT	Utilities	East	75603	73056	Bank Transfer	TXN100010
2022-01-21 00:00:00	Finance	Utilities	Central	85399	109122	Cash	TXN100011
2023-08-14 00:00:00	Sales	Marketing	Central	68086	70502	Bank Transfer	TXN100012
2021-10-04 00:00:00	HR	Training	East	99145	51102	UPI	TXN100013

Preview of the financial Performance budget dataset.

Data Validation and Preparation

Before starting the analysis, the financial dataset was reviewed and validated to make sure it was ready for further analysis using SQL, Excel, and Tableau.

The following activities were performed:

- Checked the dataset structure, column names, and data types.
- Verified the transaction date and financial amount fields.
- Checked for missing and inconsistent values.
- Reviewed department, category, region, and payment method fields.
- Verified the budget and actual expense values used in the analysis.
- Prepared the dataset for analysis, KPI calculation, and dashboard creation.

Tools & Technologies Used

Microsoft Excel

Microsoft Excel was used for KPI calculations, descriptive statistical analysis, Pivot Tables, Pivot Charts, and creating the final financial performance dashboard.

Oracle SQL

Oracle SQL was used to query and analyze the financial transaction data and generate outputs for the defined business requirements.

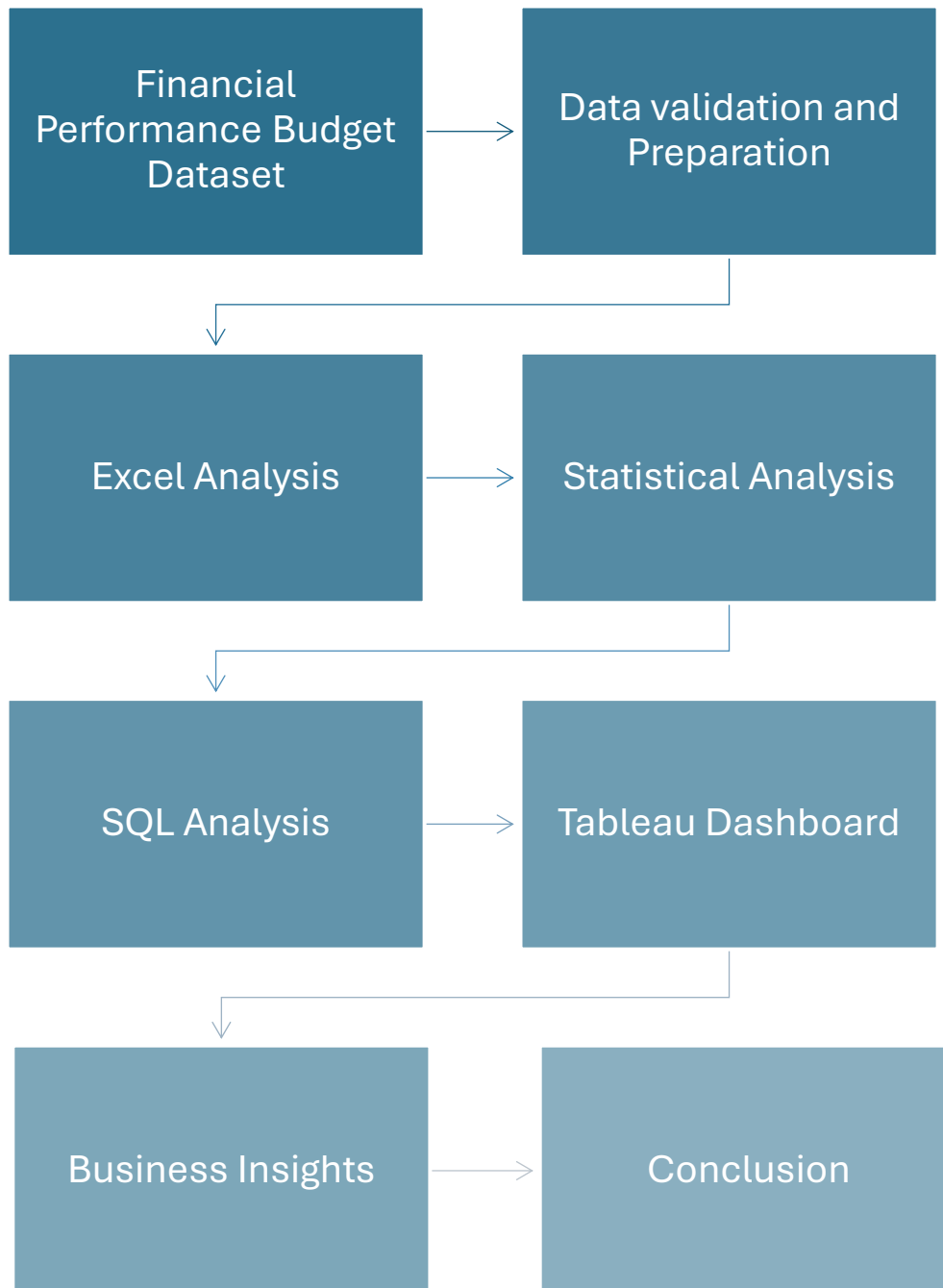
Tableau

Tableau was used to create the interactive financial performance dashboard and present budget and expense analysis through visualizations and filters.

Tableau Public

Tableau Public was used to publish the final Tableau dashboard for portfolio presentation.

Project Workflow



Excel Analysis

Data Preparation

The financial data was imported into Excel and checked before starting the analysis. I reviewed the main fields such as transaction date, department, category, region, budget amount, and actual amount to make sure the data was in a usable format. The data was then organized so that Pivot Tables, charts, and KPI calculations could be created easily.

KPI Calculations

The key financial KPIs were calculated in Excel to get a quick view of the company's overall budget and expense performance. The analysis covers Total Budget, Total Actual Expenses, Budget Variance, Budget Utilization %, and Average Expense per Transaction.

KPI Analysis				
Total Budget				79,54,69,256
Total Actual Expenses				89,02,67,029
Budget Variance				-9,47,97,773
Budget Utilization %				112%
Average Expense per Transaction				89,027

Statistical Analysis

Descriptive statistics were calculated in Excel to get a better understanding of the transaction expenses. The analysis looked at the mean, median, minimum, maximum, standard deviation, and variance to understand the typical expense and how much the transaction values varied.

Statistical Analysis Output

Statistics Analysis			
1: Mean			
		89,032.71	
2: Median			
		89,187.00	
3: Minimum			
		8,002.00	
4: Maximum			
		1,69,973.00	
5: Standard Deviation			
		46,956.17	
6: Variance			
		2,20,48,81,681.98	

Observation

The mean and median are quite close, while the difference between the minimum and maximum values shows that transaction expenses vary considerably.

Business Requirements

1 – Budget vs Actual Performance

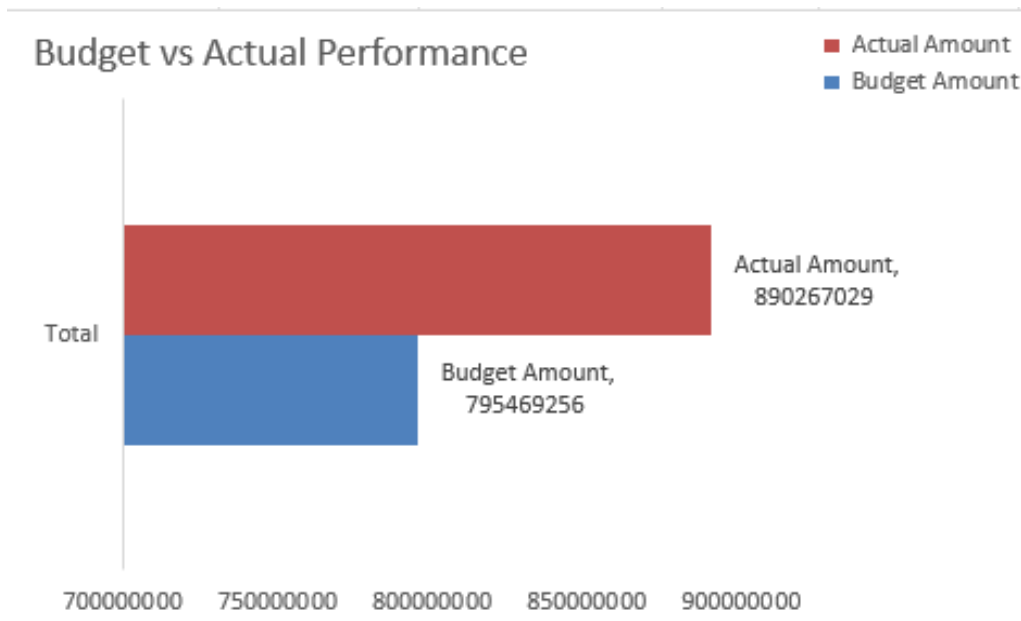
Business Requirement:

To compare the total budget with the actual expenses and identify any overall spending difference.

Pivot Table

Budget Amount	Actual Amount
795469256	890267029

Pivot Chart



Observation

Actual expenses exceeded the planned budget, indicating an overall overspend of about ₹9.48 Cr.

2 – Monthly Budget vs Actual Expenses

Business Requirement

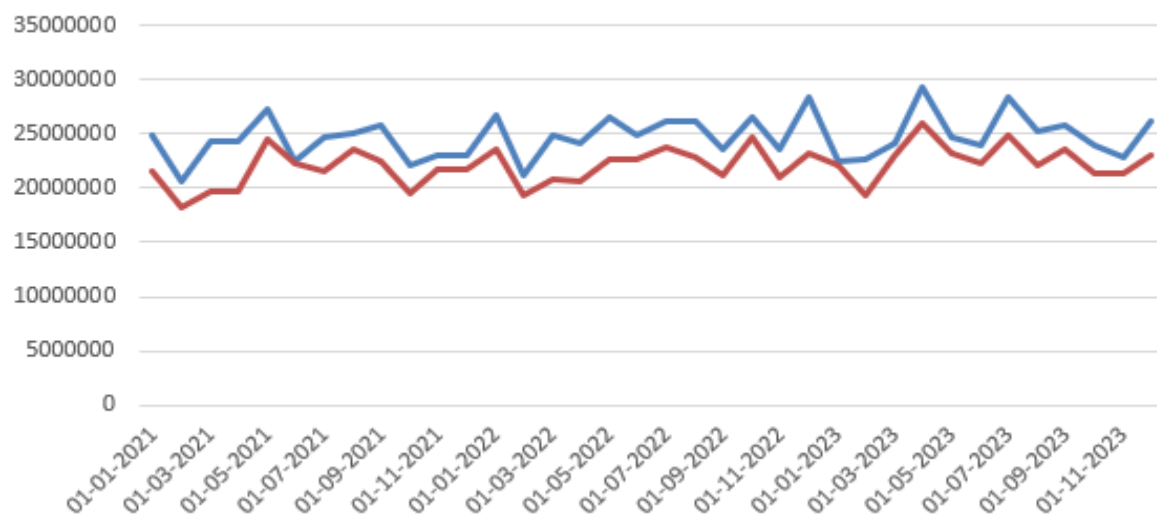
To compare monthly budgeted expenses with actual expenses and identify spending changes over time.

Pivot Table

Row Labels	Sum of Actual Amount	Sum of Budget Amount
01-01-2021	24913213	21587408
01-02-2021	20632263	18142623
01-03-2021	24295909	19747700
01-04-2021	24364104	19710246
01-05-2021	27250381	24597175
01-06-2021	22396228	22265011
01-07-2021	24685975	21471995
01-08-2021	25003789	23607929
01-09-2021	25829986	22494061
01-10-2021	22180032	19594167

Pivot Chart

Monthly Actual Expense Trend



Observation

Actual expenses remained above the budget in most months, showing a consistent spending gap.

3 – Department-wise Budget vs Actual Expenses

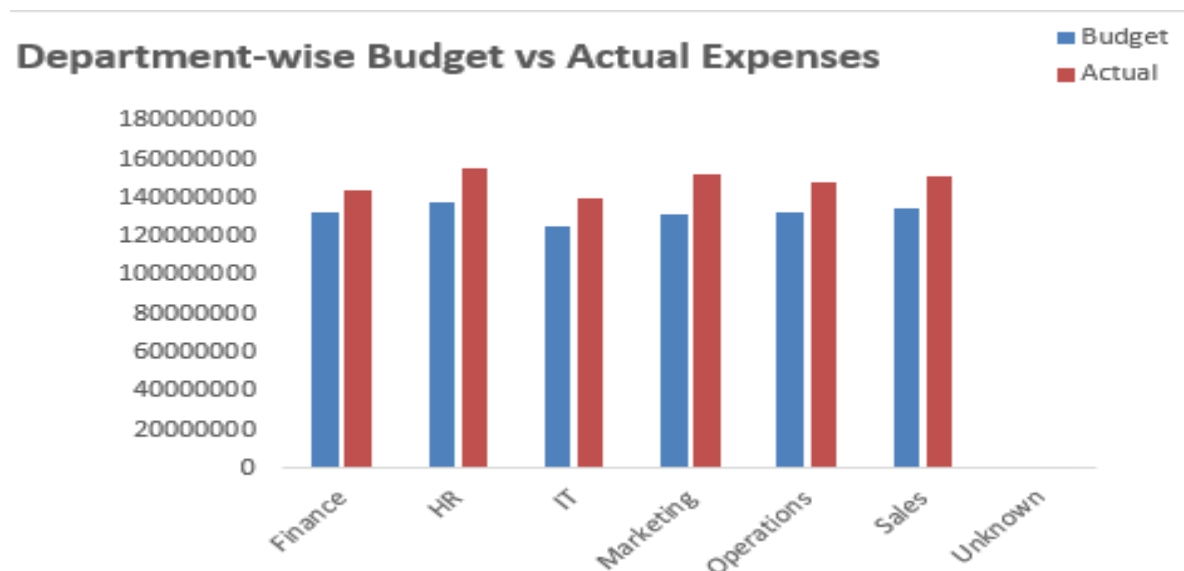
Business Requirement

To compare department-wise budgeted expenses with actual expenses and identify departments with higher spending.

Pivot Table

Row Labels	Budget	Actual
Finance	132947149	144084529
HR	137371163	155439763
IT	125771001	139663026
Marketing	131939479	151816098
Operations	132622213	147803987
Sales	134702488	151344999
Unknown	115763	114627

Pivot Chart



Observation

HR recorded the highest actual expense among the departments, while most departments exceeded their planned budgets.

4 – Category-wise Actual Expenses

Business Requirement

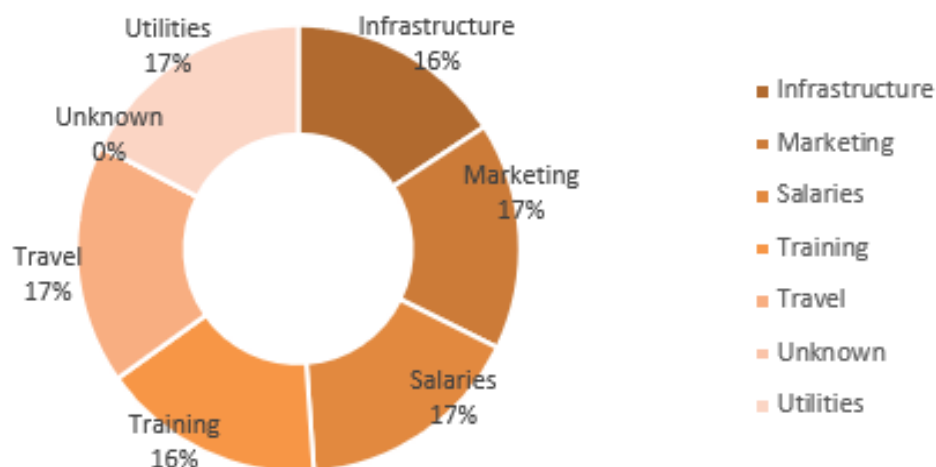
To analyze actual expenses across different categories and identify the major areas of spending.

Pivot Table

Row Labels	Actual Amount
Infrastructure	140825422
Marketing	146525696
Salaries	147744247
Training	145010926
Travel	154885333
Unknown	15085
Utilities	155260320

Pivot Chart

Category-wise Actual Expenses



Observation

Utilities recorded the highest actual expense, followed by Travel and Infrastructure.

5 – Region-wise Actual Expenses

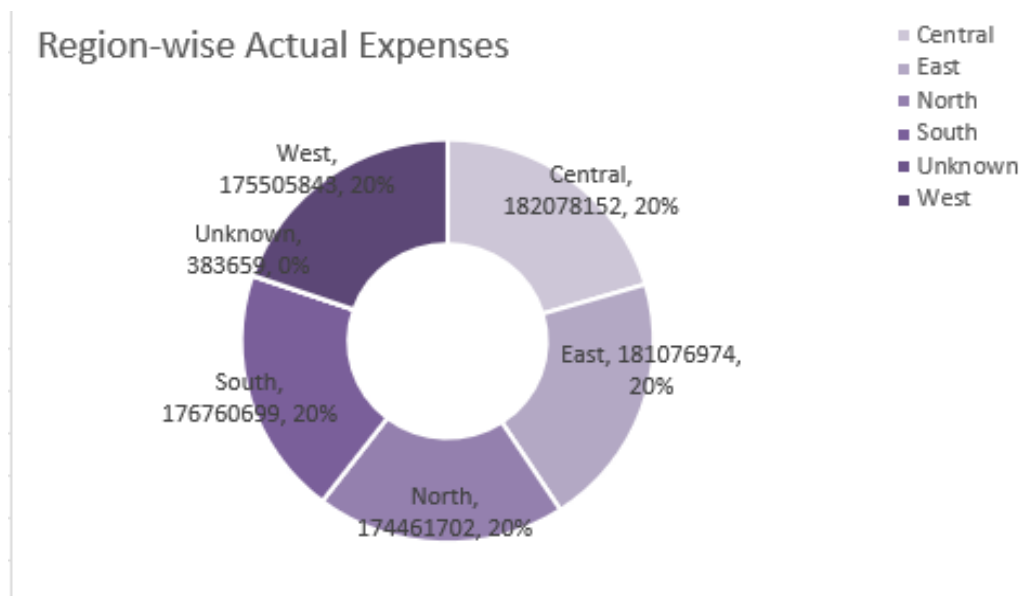
Business Requirement

To analyze actual expenses across different regions and identify the major areas of regional spending.

Pivot Table

Row Labels	Sum of Actual Am
Central	182078152
East	181076974
North	174461702
South	176760699
Unknown	383659
West	175505843

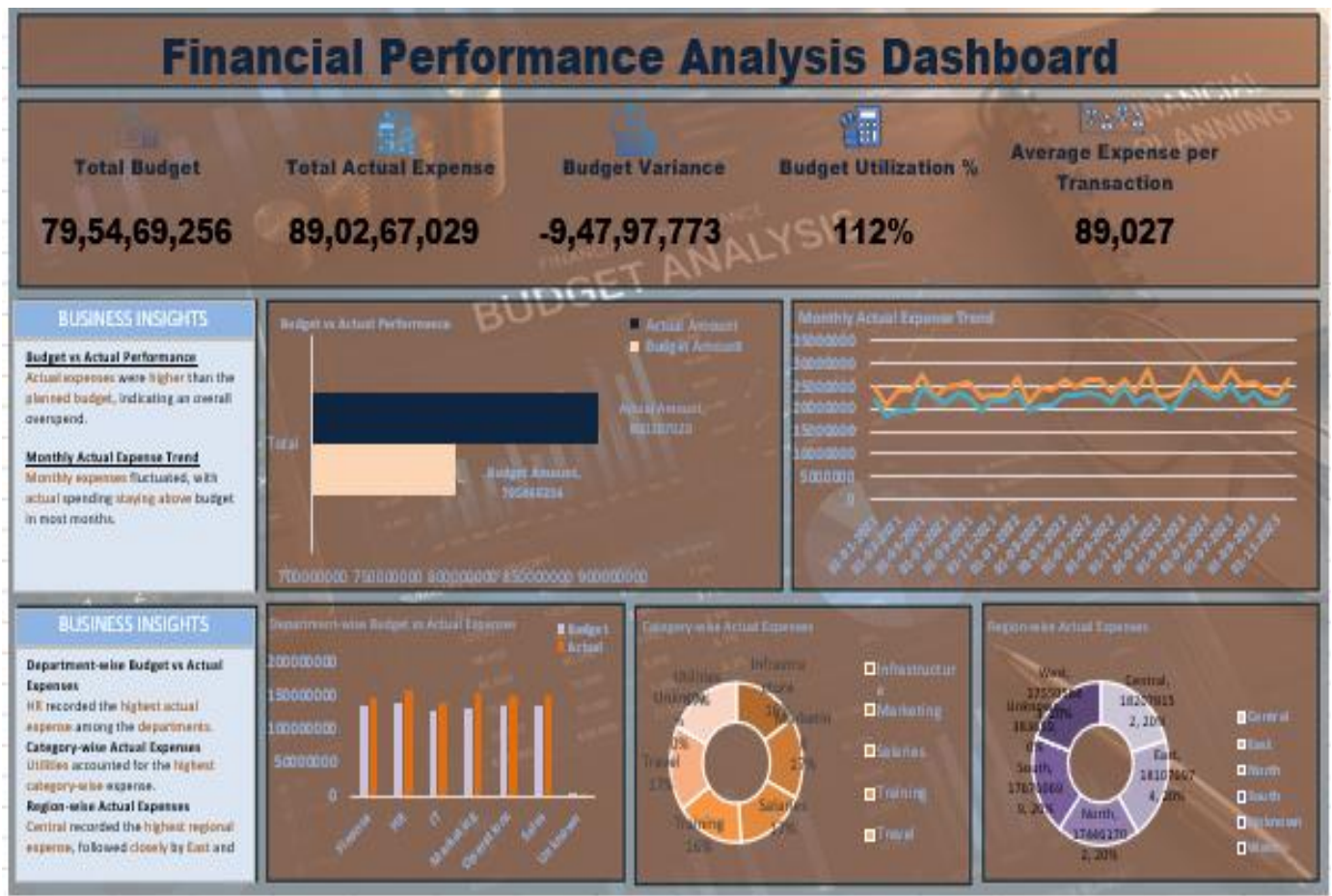
Pivot Chart



Observation

Central recorded the highest regional expense, followed closely by East and South.

Excel Dashboard



Dashboard Description

The Financial Performance Analysis Dashboard provides a clear view of the company's financial performance by comparing budgeted expenses with actual spending. It highlights key financial measures, monthly spending trends, and expense patterns across departments, categories, and regions. Date Range and Category filters make it easier to explore the data based on different selections.

Dashboard Features

- Provides a clear view of overall financial performance against the planned budget.
- Highlights monthly changes in spending and budget performance.
- Shows which departments contribute most to total expenses and where budgets are exceeded.
- Breaks down actual expenses by category and region to identify major spending areas.
- Allows users to explore financial performance interactively using Date Range and Category filters.

Key Business Insights

- Actual expenses exceeded the planned budget, indicating an overall overspend.
- Monthly expenses fluctuated, with actual spending staying above budget in most months.
- HR recorded the highest actual expense among the departments.
- Utilities accounted for the highest category-wise expense.
- Central recorded the highest regional expense.

SQL Analysis

Key Performance Indicators (KPIs)

KPI 1 – Total Budget

Business Requirement

To understand the overall budget allocated during the analysis period.

SQL Query

```
SELECT SUM(BUDGET_AMOUNT) AS TOTAL_BUDGET  
FROM FINANCIAL_DATA;
```

Output

	TOTAL_BUDGET
1	795469256

Observation

The organization planned a total budget of ₹79.55 crore during the analysis period, giving us the overall spending limit set for the business.

KPI 2 – Total Actual Expense

Business Requirement

To find out how much the organization actually spent during the analysis period.

SQL Query

```
SELECT SUM(ACTUAL_AMOUNT) AS TOTAL_ACTUAL_EXPENSE  
FROM FINANCIAL_DATA;
```

Output

	TOTAL_ACTUAL_EXPENSE
1	890267029

Observation

The organization spent ₹89.03 crore during the analysis period, which is higher than the planned budget and indicates overall overspending.

KPI 3 – Budget Variance

Business Requirement

To understand the difference between the planned budget and the actual amount spent.

SQL Query

```
SELECT SUM(BUDGET_AMOUNT) - SUM(ACTUAL_AMOUNT) AS  
BUDGET_VARIANCE  
FROM FINANCIAL_DATA;
```

Output

	BUDGET_VARIANCE
1	-94797773

Observation

The budget variance is –₹9.48 crore, showing that actual spending exceeded the planned budget by around ₹9.48 crore.

KPI 4 – Budget Utilization %

Business Requirement

To understand what percentage of the allocated budget was actually spent.

SQL Query

```
SELECT (SUM(ACTUAL_AMOUNT) / SUM(BUDGET_AMOUNT)) * 100 AS  
BUDGET_UTILIZATION  
  
FROM FINANCIAL_DATA;
```

Output

	BUDGET_UTILIZATION
1	111.92

Observation

The organization utilized 111.92% of its allocated budget, indicating that actual spending went beyond the planned budget.

KPI 5 – Average Expense per Transaction

Business Requirement

To understand the average amount spent on each transaction.

SQL Query

```
SELECT ROUND(AVG(ACTUAL_AMOUNT), 2) AS  
AVG_EXPENSE_PER_TRANSACTION  
  
FROM FINANCIAL_DATA;
```

Output

	AVG_EXPENSE_PER_TRANSACTION
1	89026.7

Observation

On average, the organization spent around ₹89,026.70 per transaction, giving a clear picture of the typical expense amount.

BUSINESS QUESTIONS

1 – Budget vs Actual Performance

Business Requirement

To compare the planned budget with actual spending and identify whether the organization stayed within its budget.

SQL Query

```
SELECT
    SUM(BUDGET_AMOUNT) AS TOTAL_BUDGET,
    SUM(ACTUAL_AMOUNT) AS TOTAL_ACTUAL_EXPENSE,
    SUM(ACTUAL_AMOUNT) - SUM(BUDGET_AMOUNT) AS
VARIANCE
FROM FINANCIAL_DATA;
```

Output

	TOTAL_BUDGET	TOTAL_ACTUAL_EXPENSE	VARIANCE
1	795469256	890267029	94797773

Observation

Actual spending was ₹94.80 million higher than the planned budget, showing that the organization spent more than originally allocated.

2 – Department-wise Budget Performance

Business Requirement

To compare budget and actual spending across departments and identify departments with higher spending.

SQL Query

```
SELECT
    DEPARTMENT,
    SUM(BUDGET_AMOUNT) AS TOTAL_BUDGET,
    SUM(ACTUAL_AMOUNT) AS TOTAL_ACTUAL_EXPENSE,
    SUM(ACTUAL_AMOUNT) - SUM(BUDGET_AMOUNT) AS VARIANCE
FROM FINANCIAL_DATA
GROUP BY DEPARTMENT
ORDER BY VARIANCE DESC;
```

Output

	DEPARTMENT	TOTAL_BUDGET	TOTAL_ACTUAL_EXPENSE	VARIANCE
1	Marketing	131939479	151816098	19876619
2	HR	137371163	155439763	18068600
3	Sales	134702488	151344999	16642511
4	Operations	132622213	147803987	15181774
5	IT	125771001	139663026	13892025
6	Finance	132947149	144084529	11137380
7	Unknown	115763	114627	-1136

Observation

Marketing had the highest overspending, exceeding its budget by about ₹19.82 million, while Finance had the lowest overspending among the main departments.

3 – Category-wise Spending Analysis

Business Requirement

To analyze actual spending across different categories and identify the categories with the highest expenses.

SQL Query

```
SELECT
    CATEGORY,
    SUM(ACTUAL_AMOUNT) AS TOTAL_ACTUAL_EXPENSE
FROM FINANCIAL_DATA
GROUP BY CATEGORY
ORDER BY TOTAL_ACTUAL_EXPENSE DESC;
```

Output

	CATEGORY	TOTAL_ACTUAL_EXPENSE
1	Utilities	155260320
2	Travel	154885333
3	Salaries	147744247
4	Marketing	146525696
5	Training	145010926
6	Infrastructure	140825422
7	Unknown	15085

Observation

Utilities had the highest spending, followed closely by Travel, while the Unknown category had very little spending.

4 – Region-wise Spending Analysis

Business Requirement

To compare actual spending across regions and identify which region has the highest expenses.

SQL Query

```
SELECT  
    REGION,  
    SUM(ACTUAL_AMOUNT) AS TOTAL_ACTUAL_EXPENSE  
FROM FINANCIAL_DATA  
GROUP BY REGION  
ORDER BY TOTAL_ACTUAL_EXPENSE DESC;
```

Output

	REGION	TOTAL_ACTUAL_EXPENSE
1	Central	182078152
2	East	181076974
3	South	176760699
4	West	175505843
5	North	174461702
6	Unknown	383659

Observation

Central had the highest spending, while North recorded the lowest spending among the identified regions.

5– Monthly Budget vs Actual Expenses

Business Requirement

To compare monthly budgeted expenses with actual expenses and identify spending gaps over time.

SQL Query

```
SELECT
    TO_CHAR(TRUNC(TRANSACTION_DATE, 'MM'), 'Mon YYYY') AS
    MONTH,
    ROUND(SUM(BUDGET_AMOUNT), 2) AS TOTAL_BUDGET,
    ROUND(SUM(ACTUAL_AMOUNT), 2) AS TOTAL_ACTUAL_EXPENSE
FROM FINANCIAL_DATA
GROUP BY TRUNC (TRANSACTION_DATE, 'MM')
ORDER BY TRUNC (TRANSACTION_DATE, 'MM');
```

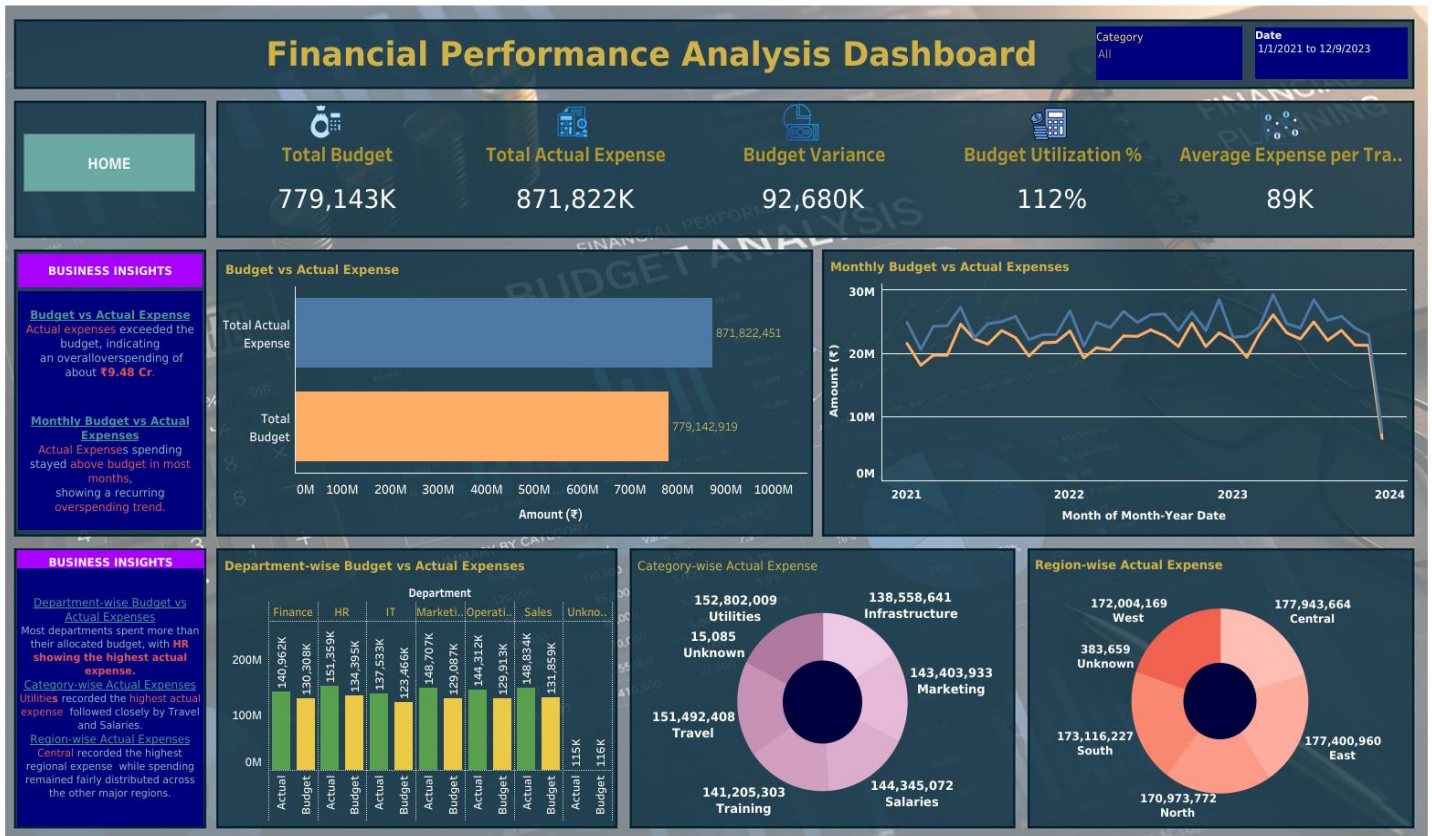
Output

	MONTH	TOTAL_BUDGET	TOTAL_ACTUAL_EXPENSE
1	Jan 2021	21587408	24913213
2	Feb 2021	18142623	20632263
3	Mar 2021	19747700	24295909
4	Apr 2021	19710246	24364104
5	May 2021	24597175	27250381
6	Jun 2021	22265011	22396228
7	Jul 2021	21471995	24685975
8	Aug 2021	23607929	25003789
9	Sep 2021	22494061	25829986
10	Oct 2021	19594167	22180032

Observation

Actual expenses were higher than the budget in most months, showing a consistent gap between planned and actual spending.

Tableau Dashboard



Description

The Financial Performance Analysis Dashboard provides an interactive view of the company's financial performance by comparing budgeted expenses with actual spending. It includes key financial indicators such as total budget, total actual expenses, budget variance, budget utilization, and average expense per transaction. The dashboard also helps users track monthly spending patterns and understand expenses across departments, categories, and regions. Interactive Date and Category filters allow users to explore the data based on their specific requirements and quickly identify areas of higher spending or budget gaps.

Dashboard Features

- Provides a clear view of overall financial performance against the planned budget.
- Highlights monthly changes in spending and budget performance.
- Shows which departments contribute most to total expenses and where budgets are exceeded.
- Breaks down actual expenses by category and region to identify major spending areas.
- Allows users to explore financial performance interactively using Date Range and Category filters.

Key Business Insights

- Actual expenses were higher than the planned budget, resulting in a budget utilization of 112%.
- Actual spending stayed above the budget in most months, showing a recurring gap between planned and actual expenses.
- HR recorded the highest actual expense among the departments, indicating a larger share of overall departmental spending.
- Utilities was the largest expense category, while the other major categories contributed fairly similar amounts.
- Central recorded the highest regional expense, with spending across the other major regions remaining relatively balanced.

Conclusion

This project successfully analyzed financial transaction data using Oracle SQL, Microsoft Excel, and Tableau. SQL was used to query and analyze the data based on the defined business requirements, while Excel was used for KPI calculations, statistical analysis, Pivot Tables, Pivot Charts, and dashboard creation. Tableau was used to build an interactive dashboard for presenting the financial performance clearly.

The analysis showed that actual expenses were higher than the planned budget, with spending patterns varying across months, departments, categories, and regions. These findings helped identify the major areas of expense and the differences between planned and actual spending, providing a clearer view of the company's financial performance.

Overall, this project provided practical experience in working with financial data, performing analysis using different tools, creating dashboards, and presenting business insights. It also helped strengthen my analytical and technical skills as an aspiring Data Analyst.

References

- Kaggle – Financial Transaction Dataset
- Oracle SQL Documentation
- Microsoft Excel Documentation
- Tableau Documentation
- Tableau Public Documentation